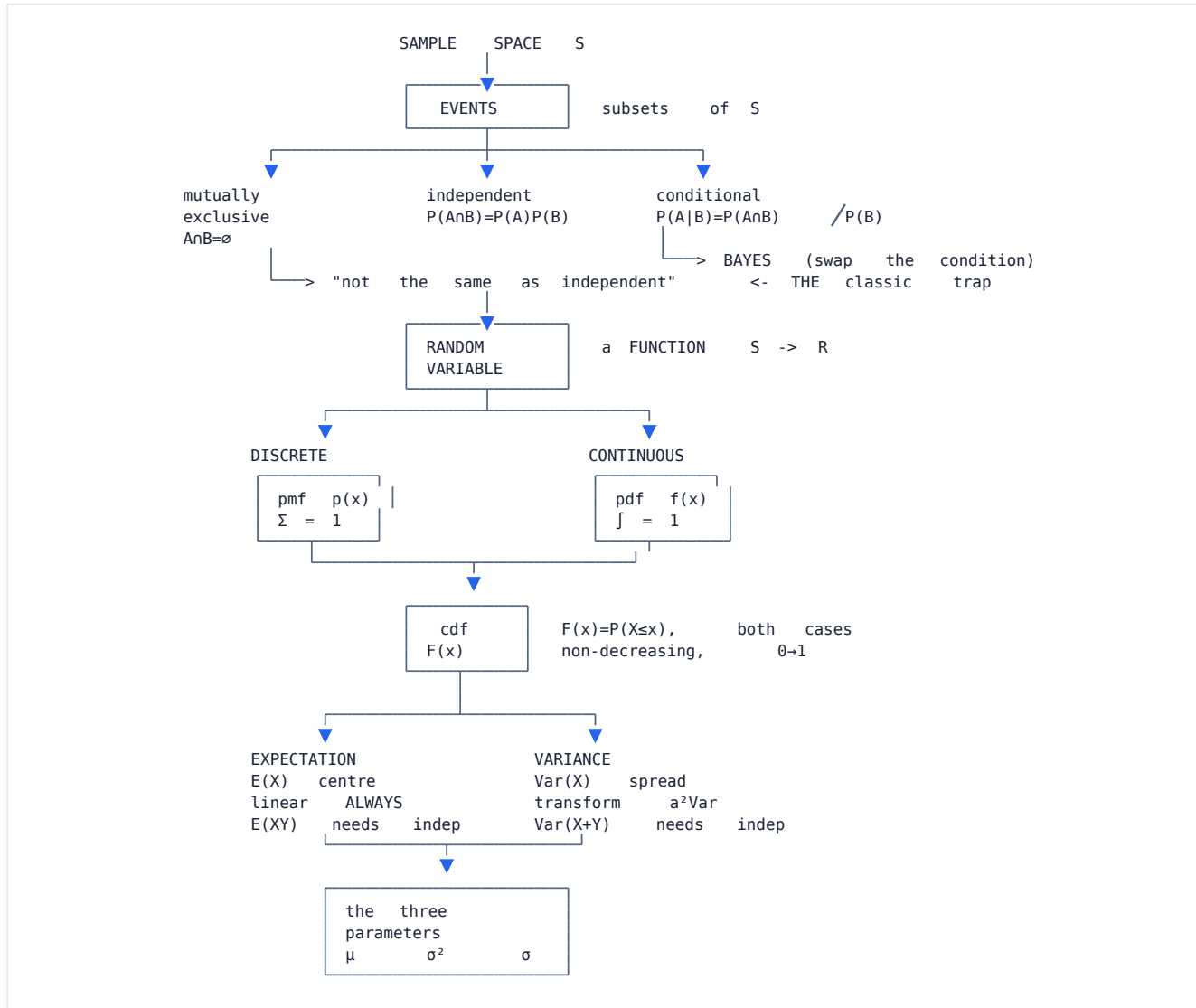


# 02 TERMS: every term, exact meaning, the trap in each

## THE TERM MAP (how the words connect)



## THE "WORDS THAT DECIDE MARKS" CHEAT

|              |                                                             |                              |
|--------------|-------------------------------------------------------------|------------------------------|
| exactly k    | → $P(X = k)$                                                | point                        |
| at least k   | → $P(X \geq k) = 1 - P(X \leq k-1)$ tail, boundary INCLUDED |                              |
| more than k  | → $P(X > k) = 1 - P(X \leq k)$                              | tail, boundary EXCLUDED      |
| at most k    | → $P(X \leq k) = F(k)$                                      | cdf                          |
| less than k  | → $P(X < k) = F(k-1)$                                       | cdf, step back one           |
| none         | → $P(X = 0)$                                                |                              |
| at least one | → $1 - P(X=0)$                                              |                              |
| expected no. | → $N \times P$                                              | (a COUNT, not a probability) |

Read with the theory. Each entry is: the term, what it exactly means, and the trap or the reason the term matters. Terms are grouped by topic.

## Probability

|                                                                                            |                                                                        |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| sample space S                                                                             | the set of all possible outcomes. TRAP: for two dice write 36 ORDERED  |
| outcomes, not 21 unordered. The counting must match the space.                             |                                                                        |
| event                                                                                      | a subset of S. TRAP: "at least one" is the complement of "none".       |
| mutually exclusive                                                                         | no two share an outcome (disjoint). TRAP: NOT the same as independent. |
| A die roll even vs prime SHARE the outcome 2, so they are not exclusive.                   |                                                                        |
| independent                                                                                | $P(A \cap B) = P(A)P(B)$ . TRAP: exclusive events with $P > 0$ are NOT |
| independent (if one happens the other cannot, so knowing A changes B).                     |                                                                        |
| conditional $P(A B)$                                                                       | the chance of A given B already happened. TRAP: the denominator is     |
| $P(B)$ , not $P(A)$ . $P(A B) \neq P(B A)$ .                                               |                                                                        |
| Bayes                                                                                      | swapping the condition: $P(A B) = P(B A)P(A)/P(B)$ . Used in the rare  |
| disease and county-forms questions.                                                        |                                                                        |
| complement discipline "at least one", "none", "rejected", "more than" are all complements. |                                                                        |
| This single move solves more probability marks than any formula.                           |                                                                        |

## Counting

|                                                                         |                                                                     |
|-------------------------------------------------------------------------|---------------------------------------------------------------------|
| permutation $P(n,r)$                                                    | ordered selections, "arrangements", "lineups", "how many ways to    |
| order". Order matters.                                                  |                                                                     |
| combination $C(n,r)$                                                    | unordered selections, "choose", "committees", "sets of". Order does |
| NOT matter. TRAP: "how many ways to form a committee of 3" is C, not P. |                                                                     |
| factorial, $0! = 1$                                                     | the seed for every count.                                           |
| repeated items                                                          | if items repeat, divide by the repeats' factorials.                 |
| complement counting                                                     | count the easy case and subtract from the total.                    |

## Random variables

|                                                                 |                                                                                         |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| random variable                                                 | a FUNCTION from the sample space to the real line. TRAP: it is not the                  |
| outcome, it is the number you attach to the outcome.            |                                                                                         |
| discrete rv                                                     | countable support (a list). Uses sums.                                                  |
| continuous rv                                                   | an interval of support. Uses integrals. $P(X = \text{one point}) = 0$ .                 |
| support                                                         | the set where the rv lives. TRAP: ALWAYS check the support before                       |
| integrating or summing; a density is zero outside it.           |                                                                                         |
| function of an rv                                               | $Y = g(X)$ . TRAP: changing $g$ changes the DISTRIBUTION of $Y$ , so recompute          |
| the support of $Y$ before anything else.                        |                                                                                         |
| Bernoulli                                                       | one trial, $X = 1$ or $0$ , $P(1)=p$ . Mean $p$ , variance $pq$ . The atom of binomial. |
| geometric                                                       | counts trials until the first success. $E = 1/p$ . Memoryless like the                  |
| exponential, in the discrete world. (no teaching slide, see 11) |                                                                                         |
| hypergeometric                                                  | draws WITHOUT replacement from a finite lot. Mean $nK/N$ , the same as the              |
| binomial mean at the same $p$ . (no teaching slide, see 11)     |                                                                                         |

## pmf, pdf, cdf

|                                                                       |                                                                         |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------|
| pmf $p(x)$                                                            | discrete, $P(X=x)$ . Must be $\geq 0$ and sum to 1.                     |
| pdf $f(x)$                                                            | continuous, a DENSITY, not a probability. $f(x)$ can exceed 1 (example: |
| uniform on width 0.5 has $f=2$ ). Only its INTEGRAL is a probability. |                                                                         |
| cdf $F(x)$                                                            | $P(X \leq x)$ for both. Non decreasing, 0 to 1, right continuous.       |
| pmf from cdf                                                          | $p(x) = F(x) - F(x-)$ , the jump. For integer rv, $F(b)-F(a-1)$ .       |
| pdf from cdf                                                          | $f = F'$ . Integrate to go up, differentiate to come down.              |
| "valid pdf" check                                                     | integrate over its support and it must equal 1. Finding $k$ is exactly  |
| this check rearranged.                                                |                                                                         |

## Expectation

|                                                 |                                                                                        |
|-------------------------------------------------|----------------------------------------------------------------------------------------|
| expectation $E(X)$                              | the weighted average, the long run mean. Symmetry trick: for symmetric                 |
| densities $E =$ the centre, no integral needed. |                                                                                        |
| $E[h(X)]$                                       | the mean of a TRANSFORMED rv. TRAP: $E[g(X)] \neq g(E(X))$ in general.                 |
| linearity                                       | $E(aX+b)=aE(X)+b$ , $E(X+Y)=E(X)+E(Y)$ with NO independence needed.                    |
| product rule                                    | $E(XY)=E(X)E(Y)$ needs INDEPENDENCE. TRAP: this is the classic true/false.             |
| expected count                                  | out of $N$ items, expected = $N \times P(\text{success})$ . No teaching slide, see 11. |

## Variance

|                       |                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------|
| variance              | $E[(X-\mu)^2]$ , the spread squared. Never negative.                                  |
| shortcut              | $E(X^2) - [E(X)]^2$ . Always easier than the definition.                              |
| standard deviation    | $\sqrt{\text{Var}}$ , in the ORIGINAL units. Use sd when Chebyshev k or a z-score     |
| is involved.          |                                                                                       |
| $\text{Var}(aX+b)$    | $a^2 \text{Var}(X)$ . TRAP: the shift b does NOTHING, the scale a squares.            |
| $\text{Var}(X-Y)$     | $\text{Var}(X)+\text{Var}(Y)$ . TRAP: the MINUS stays PLUS because Var is on squares. |
| $\text{Var}(\bar{X})$ | $\sigma^2/n$ . The root of every CLT standard error.                                  |

## Chebyshev

|                                               |                                                                                |
|-----------------------------------------------|--------------------------------------------------------------------------------|
| Chebyshev inequality                          | bounds a tail using only $\mu$ and $\sigma$ . Any distribution.                |
| k                                             | distance in STANDARD DEVIATIONS, $k = \text{eps}/\sigma$ . TRAP: raw units in, |
| must convert to sigma units first.            |                                                                                |
| vacuous bound                                 | $k \leq 1$ gives a bound of 1 or more, which says nothing.                     |
| distribution-free                             | no normal table, no assumption. That is why "which is valid without            |
| knowing the distribution" always points here. |                                                                                |

## Distributions

|                                                                                 |                                                                                            |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| binomial                                                                        | n independent 2-outcome trials, p constant, X counts successes. $E=np$ .                   |
| Poisson                                                                         | counts of rare events in a fixed window. The signature is $E = \text{Var} = \lambda$ .     |
| uniform                                                                         | constant density. Probability is proportional to interval LENGTH.                          |
| normal                                                                          | bell, symmetric, mean = median = mode = $\mu$ . Second parameter is VARIANCE.              |
| standard normal Z                                                               | $N(0,1)$ , the table world. Every normal question goes through Z.                          |
| phi vs F                                                                        | TWO table conventions. $\phi(z)$ is usually the area from 0 to z, $F(z)$ is                |
| the cumulative from $-\infty$ . MIXING THEM is the single biggest normal error. |                                                                                            |
| Decide which the question gives BEFORE substituting. (see 11)                   |                                                                                            |
| exponential                                                                     | waiting time, $\lambda = \text{rate}$ , mean = $1/\lambda$ . TRAP: rate vs mean confusion. |
| memoryless                                                                      | surviving s more given you survived to now = surviving s. No ageing.                       |
| Poisson as a limit                                                              | binomial with n big and p small, $\lambda = np$ . "Use the approximation" means            |
| Poisson with that $\lambda$ .                                                   |                                                                                            |

## Sampling, CLT

|                                                                              |                                                                                |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| population vs sample                                                         | N with parameters ( $\mu$ , $\sigma$ ) vs n with statistics ( $\bar{X}$ , s).  |
| parameter                                                                    | a number describing the population ( $\mu$ , $\sigma$ , p).                    |
| statistic                                                                    | a number computed from the sample ( $\bar{X}$ , S, $\hat{p}$ ).                |
| sampling distribution the distribution of a statistic over repeated samples. |                                                                                |
| standard error                                                               | the standard deviation of that sampling distribution. $SE = \sigma/\sqrt{n}$ . |
| CLT                                                                          | as n grows, the SAMPLE MEAN approaches normality under the theorem's           |
| conditions. $n \geq 30$ is a course heuristic, not a universal cutoff.       |                                                                                |
| exact vs approx                                                              | a normal population gives an exactly normal $\bar{X}$ at ANY n. This is the    |
| fallback when a question uses a small n.                                     |                                                                                |
| sqrt-n rule                                                                  | quadruple n, halve SE. The answer to "how large a sample".                     |

## Estimation

|                                                                            |                                                                             |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| estimator                                                                  | a RULE, a random variable ( $\bar{X}$ is a rule).                           |
| estimate                                                                   | the NUMBER once data is plugged in ( $\bar{X} = 205$ ms).                   |
| unbiased                                                                   | $E(\hat{\theta}) = \theta$ . Correct ON AVERAGE, not per sample.            |
| bias                                                                       | $E(\hat{\theta}) - \theta$ .                                                |
| consistent                                                                 | converges to $\theta$ as n grows. A large-sample property.                  |
| consistency test                                                           | sufficient pair: $E(T_n) \rightarrow \theta$ and $Var(T_n) \rightarrow 0$ . |
| efficient                                                                  | smallest variance AMONG THE UNBIASED. TRAP: a biased estimator can have     |
| a smaller variance and still lose, it is disqualified first.               |                                                                             |
| relative efficiency                                                        | $Var(T_2)/Var(T_1)$ . Bigger than 1 means $T_1$ is more efficient.          |
| sufficient                                                                 | given the statistic, the conditional sample distribution does not           |
| depend on $\theta$ . Neyman-Fisher factorisation is the usual test here.   |                                                                             |
| MSE                                                                        | $Var + Bias^2$ . The honest score when bias is allowed to compete.          |
| n-1                                                                        | the sample variance divides by n-1, not n, to be unbiased. The reason is    |
| that deviations are measured from $\bar{X}$ , which already used the data. |                                                                             |

## Words that are not statistics but decide marks

|                                                                                   |                                   |                         |
|-----------------------------------------------------------------------------------|-----------------------------------|-------------------------|
| "exactly"                                                                         | $P(X=x)$ , point probability      |                         |
| "at least k"                                                                      | $P(X \geq k) = 1 - P(X \leq k-1)$ |                         |
| "at most k"                                                                       | $P(X \leq k)$                     |                         |
| "more than k"                                                                     | $P(X > k) = 1 - P(X \leq k)$      | NOTE the boundary moves |
| "no" / "none"                                                                     | $P(X=0)$                          |                         |
| "between a and b" depends on $\leq$ vs $<$ , for continuous rv it does not matter |                                   |                         |
| "expected number" $N \times P$ , expected COUNT not probability                   |                                   |                         |
| "on average"                                                                      | mean                              |                         |
| "spread"/"variation" variance or sd, read which unit is wanted                    |                                   |                         |
| "consistent with" a distributional claim, an applicability test                   |                                   |                         |
| "using the approximation" Poisson with $I = np$ , not the exact binomial          |                                   |                         |